

Radiation Tests: HCPL7800 Isolation Amp (2)



- The chip was considered a suitable option for bricks in the modules of the end-caps
- The chip was retested as a part of system level tests at CHARM facility, before the start of pre-production, and **showed displacement damage at only $2.75 \times 10^{12} \text{ n/cm}^2$**
 - This chip has bipolar technology, but the results showed that this component failed at a lower dose in the mixed field of CHARM compared to our tests in TRIGA reactor
- The pre-production started using HCPL7800 (surface-mount package)
 - For using in the modules of the barrel, the through-hole package of this chip was considered, along with removing the chip from socket and replacing it during shut-downs of HL-LHC
- At the end, SI8920 passed the tests and was chosen for the design

